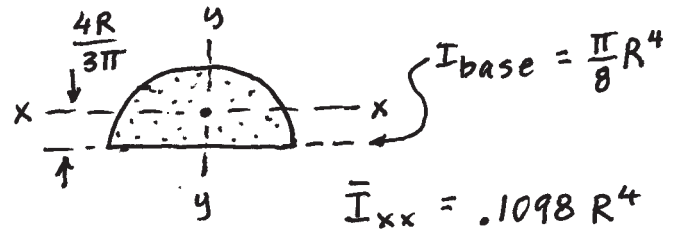
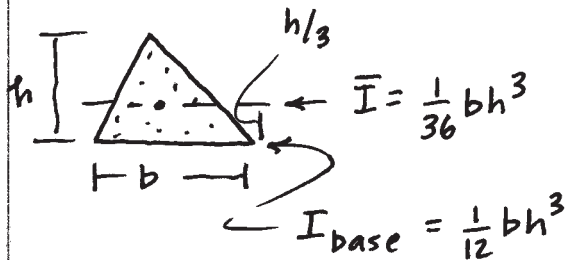
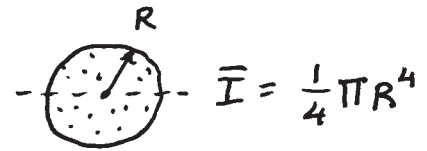
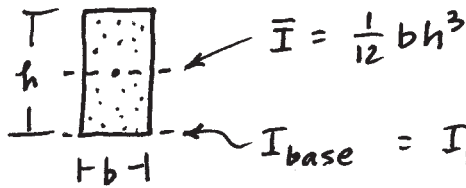


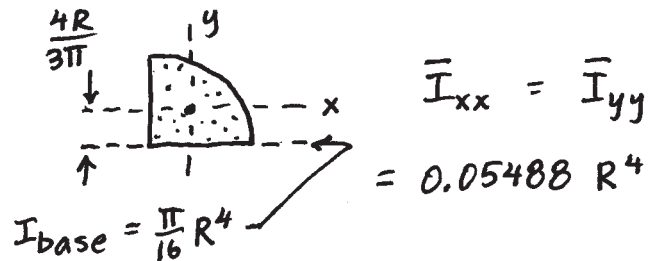
(MOI) Moments of Inertia of Common Shapes (and POI Values)



$$\bar{I}_{yy} = \frac{1}{8} \pi R^4$$

Parallel Axis Theorem (PAT):

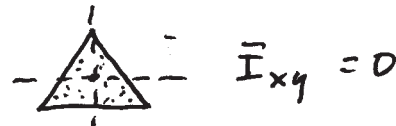
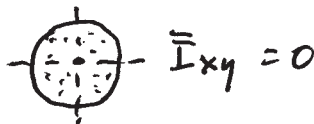
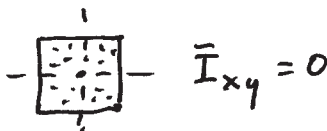
$$I = \bar{I} + Ad^2$$



(POI) Product of Inertia for Common Areas (Centroidal Values)

$\bar{I}_{xy} = 0$ for areas symmetric about either the $\bar{x}$ or $\bar{y}$ axes:

PAT: $I = \bar{I}_{xy} + A d_x d_y$



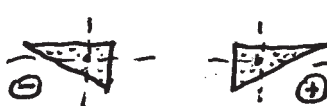
Note: I_{base} not defined for I_{xy} , bc I_{xy} is relative to a pair of orthog axes $x-y$, $x'-y'$, $u-v$

Non-Symmetric Areas (Centroidal $\bar{I}_{xy}$ Values)

Right Triangles

$n =$

Quadrant



$$\bar{I}_{xy} = (-1)^n \frac{1}{72} b^2 h^2$$

$n = \text{quadrant}$



$$\bar{I}_{xy} = (-1)^n .01647 R^4$$